Dylan McDermott

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Corteva Agriscience Research Associate (2024) | Geospatial Data Engineer | Remote Sensing | Field Sensing | Meteorologist

WORK EXPERIENCE

Meteorologist January 2025 - Present
Property & Liability Resource Bureau, Remote

- Design, build, and maintain ArcGIS-based geospatial data products used daily for weather-event verification, catastrophe analytics, and insurance-facing decision support.
- Develop automated Python and ArcPy workflows that download, quality-check, process, and update daily geospatial datasets for PLRB's ArcGIS Server, eliminating hours of manual processing and supporting reliable operational products.
- Build and maintain AWS-hosted geospatial automations and backend services using EC2, S3, RDS, and REST APIs, improving system reliability, security, and operational efficiency.
- Integrate radar, satellite, numerical-model, and observational datasets into recurring raster and vector products at regional and national scales for severe-weather, precipitation, smoke, evacuation, earthquake, and power-outage analysis.
- Contribute to an internal machine-learning project to estimate wind-gust speeds from convective thunderstorms using meteorological observations and engineered predictors.
- Advise member companies on data quality, source limitations, uncertainty, and transparent methodology for operational geospatial products.

Research Associate (Contract) June 2024 - December 2024
Corteva Agriscience, Johnston, IA
Contact: Chris Parry | Chris.parry@corteva.com

- Designed and built the Corteva "Gold Standard" automated weather station and antenna-based data relay system to transmit field measurements to the main facility in real time.
- Independently created the end-to-end Python and ArcPy pipeline for field-sensor data ingestion, quality control, spatial assignment to field and plot boundaries, statistical analysis, visualization, and research delivery.
- Helped engineer Corteva's "Smartstick," a wheeled field-sensing platform, and automated its workflow across more than seven research sites, including Johnston, Dallas Center, Lubbock, and Viluco, reducing hours of manual data processing.
- Collected weekly field data and collaborated with agronomists, engineers, and data scientists on in-canopy microclimate and field-sensing proof-of-concept work, presenting methods and findings to support continued development.
- Analyzed soil moisture, evapotranspiration, irrigation, crop-development stages, and field measurements to estimate water use and help evaluate crop stress.

ADDITIONAL EXPERIENCE

Graduate Research Assistant May 2022 - May 2024
Iowa State University, Ames, IA

- Designed and executed WRF/Noah-MP experiments to quantify how historical land-use change affected Midwest rainfall, surface fluxes, moisture transport, and mesoscale convective systems.
- Developed geospatial land-surface datasets at 3-, 15-, and 50-km resolution by translating CESM/LUMIP plant-functional-type data to IGBP/MODIS land-use classes and interpolating them to model grids.
- Processed and analyzed large gridded NetCDF datasets with Python, Xarray, and MATLAB, integrating ERA5, CESM, WRF, and observed precipitation data to produce maps, statistics, and comparative analyses.
- Communicated complex results through a 93-page thesis, technical figures, presentations, and interdisciplinary collaboration with meteorology, agronomy, engineering, and hydrology researchers.

Undergraduate Research Assistant May 2021 - August 2021
Iowa State University, Ames, IA

- Conducted boundary-layer meteorology research by identifying cases, processing atmospheric datasets, and generating data visualizations to support scientific interpretation.
- Collaborated with faculty researchers to evaluate results and communicate potential conclusions.

GEOSPATIAL & TECHNICAL SKILLS

- Geospatial data engineering and spatial analytics
- Raster and vector processing at scale
- Earth observation and remote sensing
- Coordinate systems, projections, and reprojection
- Spatial joins, buffering, interpolation, and field-level statistics
- Data ingestion, quality control, visualization, and analysis
- Weather, soil, and observational data integration
- Agricultural analytics: field boundaries, crop stages, crop stress, and irrigation
- Cloud-hosted processing, backend automation, and API integration
- Numerical weather and land-surface modeling
- Cross-functional scientific research and technical communication
- Machine-learning workflow support and feature development

SOFTWARE & DATA

- Python, SQL, MATLAB, R, C++, FORTRAN
- ArcGIS Pro, ArcPy, Enterprise, Experience Builder
- GDAL (reprojection), Rasterio, GeoPandas, Shapely
- Xarray, rioxarray, STAC APIs
- NetCDF, GeoTIFF, GRIB, Zarr
- AWS EC2, S3, RDS; REST API integration
- Sentinel-1/2, SAR, MODIS, GOES
- MRMS/radar, LiDAR, drone imagery
- WRF, Noah-MP, CESM, ERA5
- Linux, Windows, Mac, Jupyter

EDUCATION

Iowa State University, Ames, IA — January 2022 - May 2024
Master of Science, Meteorology — GPA: 3.52/4.0

- Master's Thesis: Impacts of US Deforestation on Rainfall from Mesoscale Convective Systems

Iowa State University, Ames, IA — August 2018 - May 2022
Bachelor of Science, Meteorology

- Senior Thesis: Extreme Convective Wind Magnitude and Duration in Iowa
- Dean's List honors

PERSONAL PROJECTS

AI Weather Intelligence Platform — 2025 - Present

- Architect and develop a full-stack, cloud-hosted platform that ingests, validates, processes, and delivers data from 200+ weather and Earth-observation sources through scalable Python and API workflows.
- Build production-style raster and vector workflows using Sentinel-1/2, SAR, MODIS, GOES, MRMS, numerical-model, and observational data, with automated validation and source-quality controls.

AWARDS

Esri Special Achievement in GIS (SAG) Award - PLRB — 2025

- Contributed to PLRB's award-winning use of GIS for meteorological catastrophe analysis; PLRB was one of 197 organizations selected from more than 100,000 Esri clients worldwide.